

CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

Chapter 1: Orienting Yourself: The Use of Coordinates

Topic-wise short notes with solved examples

Two numbers are enough to fix the position of any point in a plane. This chapter sets up the coordinate plane and uses it to find the distance between two points.

1 The Cartesian plane

- A **coordinate system** uses numbers to describe the exact position of a point.
- Two number lines are drawn at right angles. The horizontal line is the **x -axis** and the vertical line is the **y -axis**. Together they are the **coordinate axes**.
- The axes meet at the **origin** O , whose coordinates are $(0, 0)$.
- Equal units are marked on both axes. Distances to the **right** of O and **upwards** from O are positive. Distances to the **left** and **downwards** are negative.
- The plane containing the axes is called the **Cartesian plane**, the **coordinate plane** or the **xy -plane**.

REMEMBER

The system is named after René Descartes (1637). The negative half of each axis depends on Brahmagupta's work on zero and negative numbers (628 CE).

2 Coordinates of a point

- Every point P in the plane is written as $P(x, y)$.
- The **x -coordinate** is the perpendicular distance of P from the y -axis, measured along the x -axis.
- The **y -coordinate** is the perpendicular distance of P from the x -axis, measured along the y -axis.
- The order matters. If $x \neq y$, then $(x, y) \neq (y, x)$. The two are the same point only when $x = y$.
- A point on the x -axis has the form $(x, 0)$. A point on the y -axis has the form $(0, y)$.

Example 2.1 Write the coordinates of the point that is (a) on the x -axis, 4.5 units to the right of O (b) on the y -axis, 4.5 units below O .

SOLUTION

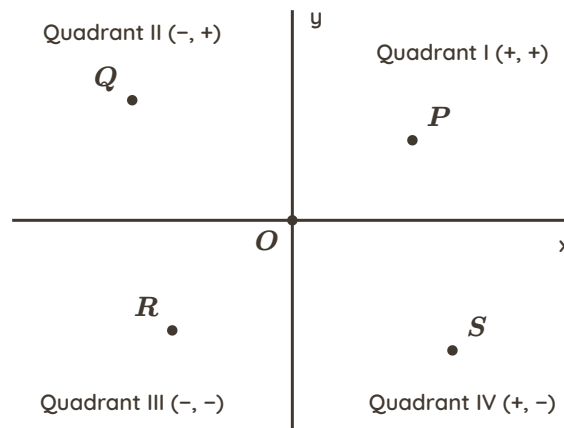
(a) It is on the x -axis, so $y = 0$. To the right means x is positive.

(b) It is on the y -axis, so $x = 0$. Below means y is negative.

(a) $(4.5, 0)$ (b) $(0, -4.5)$

3 The four quadrants

- The axes divide the plane into four parts called **quadrants**, numbered anticlockwise from the top right.



$P(3, 2)$, $Q(-4, 3)$, $R(-3, -3)$ and $S(4, -3.5)$ lie in the four quadrants

Quadrant	Sign of x	Sign of y	Example
I	+	+	$(3, 2)$
II	-	+	$(-5, 3)$
III	-	-	$(-3, -3)$
IV	+	-	$(3, -5)$

- A point on an axis does not belong to any quadrant.

Example 3.1 In which quadrant, or on which axis, does each point lie: $(-5, 3)$, $(3, -5)$, $(0, 4)$, $(-2.9, 0)$?

SOLUTION

$(-5, 3)$: x negative, y positive. $(3, -5)$: x positive, y negative.

$(0, 4)$ has $x = 0$ and $(-2.9, 0)$ has $y = 0$.

Quadrant II; Quadrant IV; on the y -axis; on the x -axis

4 Distance along a line parallel to an axis

- Two points with the same y -coordinate lie on a line parallel to the x -axis. The distance between (x_1, y) and (x_2, y) is $|x_2 - x_1|$.
- Two points with the same x -coordinate lie on a line parallel to the y -axis. The distance between (x, y_1) and (x, y_2) is $|y_2 - y_1|$.
- The bars mean the **absolute value**: take the difference and ignore its sign.

Example 4.1 A door has its two ends at $(8.5, 0)$ and $(11.5, 0)$. How wide is it? Another door has ends at $(0, 1.5)$ and $(0, 4)$. Which door is wider?

SOLUTION

First door: $|11.5 - 8.5| = 3$ units. Second door: $|4 - 1.5| = 2.5$ units.

The first door is 3 units wide and is the wider one.

Example 4.2 Three corners of a rectangular table are at $(8, 9)$, $(11, 9)$ and $(11, 7)$. Find the fourth corner and the length and width of the table.

SOLUTION

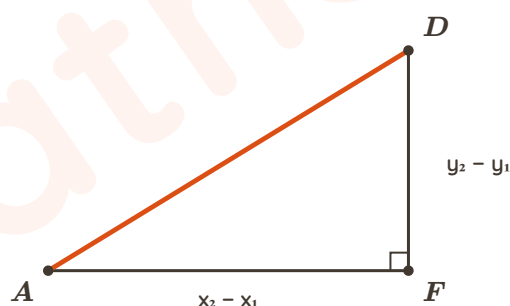
The fourth corner has the x -coordinate of the first point and the y -coordinate of the third.

Length = $|11 - 8| = 3$ units and width = $|9 - 7| = 2$ units.

Fourth corner $(8, 7)$; the table is 3 units by 2 units.

5 The distance formula

- For a slanting segment, draw a right-angled triangle with sides parallel to the axes and use the Baudhāyana-Pythagoras theorem.



AD is the hypotenuse of a right-angled triangle

- The distance between $A(x_1, y_1)$ and $D(x_2, y_2)$ is

$$AD = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- It makes no difference whether $x_2 - x_1$ and $y_2 - y_1$ are positive or negative, because they are squared.
- The distance of (x, y) from the origin is $\sqrt{x^2 + y^2}$.

Example 5.1 Find the lengths of the sides of the triangle with vertices $A(3, 4)$, $D(7, 1)$ and $M(9, 6)$.

SOLUTION

$$AD = \sqrt{(7-3)^2 + (1-4)^2} = \sqrt{16+9} = 5 \text{ units}$$

$$DM = \sqrt{(9-7)^2 + (6-1)^2} = \sqrt{4+25} = \sqrt{29} \text{ units}$$

$$MA = \sqrt{(3-9)^2 + (4-6)^2} = \sqrt{36+4} = \sqrt{40} \text{ units}$$

$$AD = 5, DM = \sqrt{29} \text{ and } MA = \sqrt{40} \text{ units}$$

Example 5.2 Show that $A(2, 1)$, $B(-1, 2)$, $C(-2, -1)$ and $D(1, -2)$ are the vertices of a square, and find its area.

SOLUTION

$AB^2 = 9 + 1 = 10$, $BC^2 = 1 + 9 = 10$, $CD^2 = 9 + 1 = 10$, $DA^2 = 1 + 9 = 10$, so all four sides are equal.

Diagonals: $AC^2 = 16 + 4 = 20$ and $BD^2 = 4 + 16 = 20$, so the diagonals are equal.

Four equal sides and equal diagonals make a square. Area = $(\sqrt{10})^2$.

$ABCD$ is a square of area 10 square units.

6 Reflection in the axes

- Reflecting a point in the **y -axis** changes the sign of its x -coordinate: $(x, y) \rightarrow (-x, y)$.
- Reflecting a point in the **x -axis** changes the sign of its y -coordinate: $(x, y) \rightarrow (x, -y)$.
- A reflection does not change lengths. A triangle and its image have equal sides.

Example 6.1 Triangle ADM with $A(3, 4)$, $D(7, 1)$, $M(9, 6)$ is reflected in the y -axis. Write the images and check that $A'D' = AD$.

SOLUTION

$$A'(-3, 4), D'(-7, 1) \text{ and } M'(-9, 6)$$

$$A'D' = \sqrt{(-7+3)^2 + (1-4)^2} = \sqrt{16+9} = 5 \text{ units, the same as } AD.$$

The reflection keeps the lengths of the sides.

7 Using distances: collinear points, mid-points and circles

- **Collinear points:** three points lie on one line if the largest of the three distances between them equals the sum of the other two.
- **Mid-point:** the mid-point M of the segment joining $S(x_1, y_1)$ and $T(x_2, y_2)$ is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Circle:** a point lies on a circle with centre C and radius r if its distance from C equals r . It lies inside if the distance is less than r and outside if it is greater.

Example 7.1 Are the points $M(-3, -4)$, $A(0, 0)$ and $G(6, 8)$ on one straight line?

SOLUTION

$$MA = \sqrt{9 + 16} = 5, AG = \sqrt{36 + 64} = 10, MG = \sqrt{81 + 144} = 15$$

$$MA + AG = 5 + 10 = 15 = MG$$

Yes, the three points are collinear.

Example 7.2 $M(-7, 1)$ is the mid-point of $A(3, -4)$ and $B(x, y)$. Find B .

SOLUTION

$$\frac{3 + x}{2} = -7 \text{ gives } x = -17, \text{ and } \frac{-4 + y}{2} = 1 \text{ gives } y = 6.$$

$$B = (-17, 6)$$

Example 7.3 Show that $A(1, -8)$, $B(-4, 7)$ and $C(-7, -4)$ lie on a circle with centre at the origin. Does $D(-5, 6)$ lie inside, on or outside this circle?

SOLUTION

$$OA = \sqrt{1 + 64} = \sqrt{65}, OB = \sqrt{16 + 49} = \sqrt{65}, OC = \sqrt{49 + 16} = \sqrt{65}$$

All three are at the same distance from O , so they lie on a circle of radius $\sqrt{65}$.

$$OD = \sqrt{25 + 36} = \sqrt{61}, \text{ which is less than } \sqrt{65}.$$

The radius is $\sqrt{65}$ units, and D lies inside the circle.

8 Common mistakes

- Writing the coordinates in the wrong order. The x -coordinate always comes first.
- Thinking a point on an axis is in a quadrant. It is on the boundary and belongs to none.
- Sign errors with negative coordinates: $-3 - (-7) = 4$, not -10 .
- Forgetting to square before adding in the distance formula, or forgetting the square root at the end.
- Saying three points are collinear because two of the distances are equal. Check that two distances **add up** to the third.